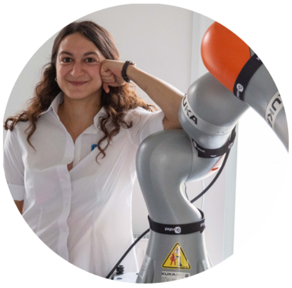
Daniella Tola



Melbourne, VIC

daniellatola.au

Citizenship:

Australian and Danish

Languages:

English (mother tongue)

Danish (fluent)

Assyrian (proficient)

Arabic (conversational)

Methods & Approaches:

MBSE:
UML, SySML, co-simulation

Safety Engineering:
FTA, Risk matrix, GSN

Requirements Engineering:
Specification, Traceability matrix, Acceptance tests

Summary

Senior software engineer with a PhD in robotic systems integration and hands-on experience developing ML, computer vision, and ROS2 systems in manufacturing. Excels at systems thinking by rapidly understanding complex technical and operational challenges across diverse domains (manufacturing, pharmaceuticals, agriculture) to architect innovative solutions. Technical expertise spans algorithm development, hardware-software integration, and deploying models on edge devices using Python. Demonstrated leadership through mentoring teams, driving technical decisions, proposing innovative solutions (including patent-assessed work), and building strong client relationships. Additional depth in safety engineering, digital twins, and systems modelling, supporting end-to-end understanding from architecture through to deployment.

Relevant Work Experience

2026/06 - Present, Senior Data Scientist at Sparx Systems

Key Responsibilities:

- Develop AI-based coding assistant for internal use

Achievements:

- Designed and delivered three LLM-powered features integrated into an internal web UI within the first two weeks of onboarding, exceeding the initial goal of one functional feature
- Architected a multi-strategy interaction pattern for AI assistant features, covering context menu, dialogue, and ribbon/browser entry points, to support diverse user workflows
- Documented workflow diagrams to support knowledge transfer within the team
- Proactively modularised and refactored Python code to improve maintainability and development velocity, despite this being outside the initial scope
- Engineered LLM prompts and structured output parsing pipelines in Python

2026/04 - 2026/06, Senior Robotics & ML Engineer (Senior Data Scientist) at Trifork
2024/10 - 2026/03, Robotics & ML Engineer (Data Scientist) at Trifork

Key Responsibilities:

- Develop ML and computer vision production code for object detection in different domains: train tracks and signals, manufacturing
- Design and implement ROS2 nodes and software architecture decisions
- Interface software with industrial camera systems and edge computing hardware (Jetson)
- Implement and support software and ML pipeline: annotate, train, CI/CD, and deploy

Achievements:

- Proposed novel computer vision solutions within first week of current project, one assessed for patent potential by client
- Identified data leakage and distribution issues through systematic dataset quality assessment, preventing overestimated model performance ahead of deployment
- Worked across the full stack from algorithm development through deployment in fast-paced prototyping environment
- Built strong client relationships resulting in direct client request for continued project engagement despite relocating from Denmark to Australia
- Proposed and implemented peer review processes for ML models across the department, improving cross-team consistency

2024/04 - 2024/09, Postdoc at Aalborg University with Novo Nordisk

Key Responsibilities:

- Conceptualise novel aseptic factory layouts for optimising low volume small batch production
- Conduct interviews with domain experts to inform design decisions
- Analyse complex requirements including cleanroom standards (cleanroom class A), low-volume specialised production, and cost optimisation

Achievements:

- Broke down process models conceptually using Enterprise Architect, and traced components back to requirements
- Generated innovative layout concepts by drawing insights from diverse unrelated domains

2025/01 - 2025/12 (Part-time), Assistant Lecturer at Aarhus University

Redesigned lecture materials on Linear Algebra with real-world examples and step-by-step explanations, resulting in positive student feedback. Supervised 2 student group projects.

Additional Work Experience

- **2023/11 - 2024/03, Postdoc at Aarhus University**
  Developed AR applications in a Digital Twin context. Further developed software tool for exporting FMUs in Python and C#.

- **2020/08 - 2020/11, Research Assistant at Aarhus University**
  Developed dataset for fault detection in screwdriving and classified faults using ML.

- **2020/02 - 2020/06 (Part-time), Teaching Assistant at Aarhus University**
  Stochastic Modelling & Processing and Network Programming & Basic Communication.

- **2019/07 - 2019/09, Student Programmer at Aarhus Universitetshospital**
  Simulated system for liquid control in human brains, and performed *in vitro* experiments.

- **2017/02 - 2019/09 (Part-time), Mentor for students at Aarhus University**
  Helped students with diverse learning needs, providing individualised support and coaching on specialised learning techniques and study strategies.

- **2018/09 - 2018/11 (Part-time), Robotics Developer at Capra Robotics**
  Various tasks related to robotics development such as programming in ROS, control, implementing communication protocols, and tasks related to project and risk management.

- **2017/01 - 2017/06, Software Development Intern at Marel**
  Refactored object detection image processing algorithm using C++, visualised the configuration of a delta robot on a webpage using Three.js.

Education

2020/11 - 2023/10, Ph.D. in Robotics with Danish Company Technicon, Aarhus University

Study abroad period (September-December 2022) at Queensland University of Technology in Australia with Distinguished Professor Peter Corke. The Ph.D. was in collaboration with the Danish system integrator, Technicon, on optimising **robotic systems integration** processes via a robotic systems configurator, developing digital shadows of robotic systems, and researching the de-facto standard robot modelling format, URDF.

2018/08 - 2020/06, M.Sc. in Computer Engineering, Aarhus University

Student exchange period (September-January 2019) at Katholieke Universiteit Leuven in Belgium with focus on automation and control. Specialisation in Machine Learning, Computer Vision, and Embedded Devices. Thesis on agricultural machinery safety, included safety engineering analysis (FTA, risk matrix, GSN safety cases) and co-simulation using FMI-compliant interfaces modelled in VDM. Finished with a GPA of 11.2/12.0.

2015/02 - 2018/06, B.Eng. in Electronics, Aarhus University

Thesis on building a vertical farming cabinet system. Finished with a GPA of 10.5/12.0.

Selected Technical Skills

Systems Engineering and Modelling

- Modelling languages: UML, SysML, VDM (formal specification)
- Tools: Enterprise Architect, Visio, 20-sim, INTO-CPS
- Safety Engineering: FTA, Risk matrix, GSN, Hazard analysis
- Co-simulation: FMI/FMU standard, Digital Twins

Programming Languages

- C
- C++
- C#
- Python
- MATLAB
- VDM (Formal modelling)

Robotics and Visualisation

- ROS/ROS2 (node development, real-time systems)
- Sensor integration (mainly cameras)
- Embedded systems and edge computing (NVIDIA Jetson)
- Visualisation and Simulation:
  - Gazebo
  - URSim
  - URDF
  - Unity
  - Blender

Machine Learning and Computer Vision

- Current focus: Object detection models including YOLO, TAO, and MMDetection
- Annotation: Darwin, LabelStudio, MVTec Deep Learning Tool
- Python libraries: pandas, polars, ultralytics, roboflow, sklearn, opencv, matplotlib, keras

Development Tools

- Google Cloud
- UpCloud
- Git
- Docker
- Atlassian
- CI/CD
- Tailscale

Publications

Selected From 10+ Publications (View Full List)

- D. Tola and P. Corke, "Understanding URDF: A Survey Based on User Experience," in *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)*, 2023, pp. 1–7. DOI: 10.1109/CASE56687.2023.10260660
- D. Tola, E. Madsen, C. Gomes, L. Esterle, C. Schlette, C. Hansen, and P. G. Larsen, "Towards Easy Robot System Integration: Challenges and Future Directions," in *2022 IEEE/SICE International Symposium on System Integration (SII)*, 2022, pp. 77–82. DOI: 10.1109/SII52469.2022.9708846

Open Source Contributions

- **URDF Dataset:** Curated dataset of 300+ robot models from diverse sources, supporting robotics research and development. 45k+ views, 5.1k+ clones.

Selected Volunteer Experience

2026 (2 full-day workshops), Tutor at Melbourne Girls Programming Network
Taught Python to high school girls and supported with exercises and tutoring tasks.

2024/09 - 2025/08 (Part-time), Teacher at ReDI School of Digital Integration
Taught Python to women and non-binary individuals from migrant/refugee backgrounds, adapting approach for learners with no mathematical background. Included preparing course materials, teaching, and correcting assignments with feedback.

2022/12/06-08, Assistant at Australasian Conference on Robotics and Automation
Managed registrations, assisted with audiovisual setups, and helped out with the food service.

2016/11 - 2017/08 (Part-time), Tutor at "Red Barnet" (Save the Children) Ungdom
Helped middle school students with mathematics- and physics-related homework.